

Customer Churn & Retention Analysis — Summary

Telco Customer Churn Dataset | Data Analysis Internship, Task 4

Total Customers	Churned	Churn Rate	Avg Monthly Charges	Revenue at Risk
7,043	1,869	26.54%	\$64.76	\$139,130.85

Theoretical Background & Methodology

Customer Churn

Customer churn refers to the rate at which customers stop doing business with a company over a given period — here, customers who have discontinued their telecom service. Churn is one of the most closely watched metrics in subscription-based industries because acquiring a new customer typically costs far more than retaining an existing one. Understanding which factors drive churn allows a business to move from reactive win-back efforts to proactive, targeted retention.

Chi-Square Test of Independence

The chi-square test evaluates whether two categorical variables are statistically independent of one another — in this case, contract type (Month-to-month, One year, Two year) and churn status (Yes/No). It compares the observed frequency of customers in each contract-by-churn combination against the frequency that would be expected if there were no relationship between the two variables. The test statistic, χ^2 , grows larger as the gap between observed and expected counts widens. A resulting p-value below the chosen significance level ($\alpha = 0.05$) means the observed pattern is extremely unlikely to have occurred by chance alone, so the null hypothesis of independence is rejected in favor of a real association between contract type and churn.

Independent Samples T-Test

The independent samples t-test compares the mean of a continuous variable — monthly charges — between two separate groups: customers who churned and customers who did not. It asks whether the difference between the two-group means is larger than what would be expected from random sampling variation alone, given the spread (standard deviation) and size of each group. The resulting t-statistic reflects how many standard errors apart the two means are; a large $|t|$ paired with a p-value below 0.05 indicates the difference in average monthly charges between churned and retained customers is statistically significant, not a coincidence of the sample drawn.

Rule-Based Customer Segmentation

Rather than clustering algorithms, this analysis groups customers using simple, interpretable business rules based on two variables known in advance to correlate with churn: tenure (how long a customer has stayed) and monthly charges (how much they pay). Because churn is not one of the variables used to define the groups, each segment's churn rate is a genuine outcome that can be measured and compared, rather than something built into the segment definition. This keeps the segmentation transparent and directly actionable — a High-Risk customer, for instance, is identifiable the moment they sign up, before any churn behaviour has occurred.

Statistical Test Results ($\alpha = 0.05$)

Test	Statistic	P-value	Decision
Chi-Square: Contract vs Churn	$\chi^2 = 1184.60$	$p < 0.001$	Reject H0 — associated
T-Test: Monthly Charges vs Churn	$t = 18.41$	$p < 0.001$	Reject H0 — means differ

Customer Segments (rule-based on tenure & monthly charges)

Segment	Customers	Avg Tenure	Avg Monthly \$	Churn Rate
High-Risk (new + expensive)	880	5.1	\$83.00	68.07%
New (≤ 12 month, lower charge)	1,306	4.5	\$37.97	33.54%
High-Value (established, high \$)	2,711	47.0	\$92.52	24.90%
Loyal (established, lower \$)	2,146	42.0	\$38.52	7.32%

Priority segment: High-Risk — 880 customers, brand-new (avg. 5.1 months) and already on expensive plans (avg. \$83/month), churning at 68.1%, nearly 3x the overall rate.

Key Findings & Recommendations

Finding	Recommendation
Overall churn is 26.54% (1,869 of 7,043 customers).	Treat retention as a company-wide priority; track churn rate monthly as a core KPI.
Contract type is the strongest driver: Month-to-month churns at 42.71% vs 11.27% (1-yr) and 2.83% (2-yr); confirmed significant by chi-square ($p < 0.001$).	Incentivize longer contracts — discount or perk for switching Month-to-month customers to annual plans.
New customers churn most: 47.44% in months 0-12, falling to 9.51% after 48+ months.	Build a structured first-year onboarding/check-in program targeting 0–12-month customers.
Churned customers pay significantly more (\$74.44 vs \$61.27, t-test $p < 0.001$); Fiber optic churns at 41.89% vs 18.96% for DSL.	Review Fiber optic pricing/value — consider bundle perks or a loyalty rate for high-tier plans.
Electronic check payers churn at 45.29% (vs ~15-17% for automatic methods); customers without Tech Support churn at 41.64% vs 15.17% with it.	Nudge Electronic check users to automatic payment; proactively offer Tech Support, prioritizing the 880 High-Risk customers first.